

EEGSchizNet

EEG-Based Schizophrenia Screening via Multi-Branch Deep Learning & Uncertainty Quantification



1. PRODUCT OVERVIEW

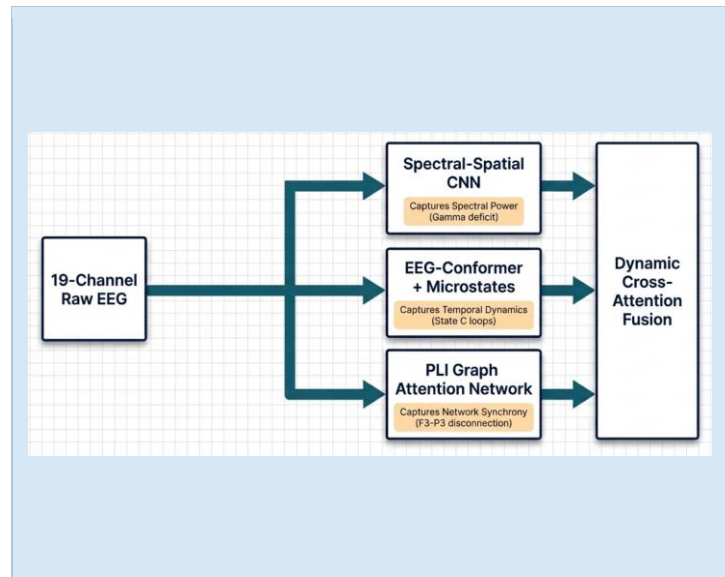
EEGSchizNet is a deep learning pipeline for schizophrenia screening from resting-state EEG, built on the REPOD dataset (28 subjects, 19-channel, 250 Hz). It fuses four parallel branches—a pretrained LaBraM transformer backbone, a Spectral CNN targeting frequency-domain biomarkers, an EEG-Conformer capturing temporal dynamics, and a PLI-GAT graph network modelling phase-based connectivity—into a single 256-dimensional fused embedding via a residual fusion gate. A Monte Carlo Dropout classifier (30 stochastic passes) produces per-subject risk scores with calibrated epistemic uncertainty, yielding three clinical risk zones (Normal / Borderline / High Risk) alongside interpretable GradCAM electrode saliency maps.

2. USE CASES & MOTIVATION

- Early schizophrenia screening in outpatient or primary care settings where specialist access is limited
- Research tool for studying EEG biomarkers of psychosis across clinical cohorts
- Augmenting psychiatrist assessment with objective, reproducible neural risk scores
- Validation platform for novel EEG-based biomarkers (PLI connectivity, LZ complexity, theta-gamma MI)

3. TECHNICAL PIPELINE

- REPOD EDF preprocessing: bandpass filter (0.5–45 Hz), epoch extraction (4 s windows), 250 Hz native sampling
- Parallel branch extraction: Spectral CNN features (256-dim),
- EEG-Conformer embeddings (256-dim),
- PLI-GAT graph embeddings (256-dim)
- Fusion Gate: three branch outputs → single 256-dim embedding
- Training: 5-fold subject-stratified cross-validation, asymmetric loss (SCZ weight = 2.5), AdamW, early stopping (patience = 15)
- Evaluation: MC Dropout inference, five biomarker profiles, GradCAM topoplots, automated per-subject clinical PDF report



4. TARGET AUDIENCE & CUSTOMERS

Clinical & Diagnostic

- Psychiatrists and neurologists requiring objective EEG-based decision support
- Clinical neuroscience researchers studying schizophrenia biomarkers
- Hospital EEG labs and neurodiagnostic centres
- Academic institutions and mental health research groups

5. UNIQUE VALUE PROPOSITION

EEGSchizNet is the only open research prototype combining three biologically motivated signal branches, and uncertainty-quantified MC Dropout output into a single end-to-end schizophrenia screening pipeline with interpretable clinical reports.

Advantage

Why It Matters

Multi-Branch Biological Coverage	Captures spectral, temporal, and graph-based EEG signatures simultaneously — none of which alone fully characterises schizophrenia — giving clinicians multi-dimensional evidence per patient.
Calibrated Uncertainty Quantification	Unlike binary yes/no predictions, MC Dropout surfaces when the model is uncertain, flagging borderline cases for mandatory clinical review rather than auto-classifying them.
Interpretable Clinical Output	GradCAM electrode saliency maps and five validated biomarker profiles (Beta Power Ratio, Theta/Alpha, PSI, Theta-Gamma MI, LZ Complexity) translate neural network decisions into neuroscientifically interpretable evidence.